AFDB – DATA INNOVATION FOR AFRICA (DI4A)

A price index built from the web

Three notebooks, eight countries, two families of sources: e-commerce and official publications.

68,884
products

53,149
official records

277
tests

Figures measured on 2026-10-28 at 12:10

AGENDA

What this presentation covers

1. **The objective**
   what we are trying to measure, and what we are not
2. **The concepts**
   COICOP, basket, variety, SKU, matched models, weights
3. **The chain**
   the diagram of the whole device, from sites to index
4. **The three notebooks**
   purpose, sources, outputs; classification in detail
5. **The calculations**
   each measure walked through on a real example
6. **Running it**
   install, launch, resume an interrupted collection
7. **The data**
   what has been collected, country by country
8. **Scope and limits**
   what the device is worth, and where it stops

The problem

A consumer price index is collected in the field: enumerators record prices in outlets, **every month**.

It is slow, costly, and covers only what we go and see.

THE QUESTION

The web publishes prices continuously, free and in bulk. **What can be drawn from it for official statistics, and under what conditions?**

This device answers by building the whole chain — collection, classification, index — across **eight African countries**, and by measuring what it is worth.

What we produce, what we do not

PRODUCED

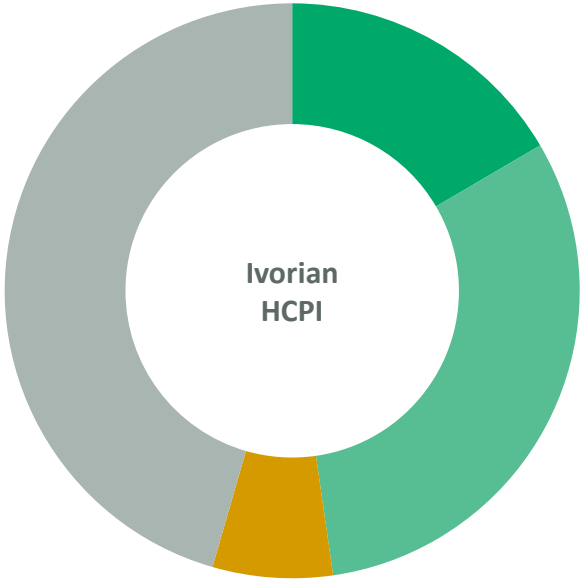
- time-stamped price records, by product and by country
- a COICOP 2018 classification of every product
- elementary and aggregate indices, several formulas
- a reproducible archive, with its collection log
- the share of the official basket actually observed

NOT PRODUCED

- a consumer price index: half the basket is missing
- rents actually paid: the web publishes asking prices
- services — schooling, health, transport, catering
- fresh produce, which is not sold online
- national weights, except where the office publishes them

What share of the official basket can be seen?

Measured against the official Ivorian weights, group by group; the verdict is declared, not computed.



Goods sold online, excluding food	**16.56 %**
Food item 01.1, to be split fresh / packaged	**31.17 %**
Partly observable (car parts, household upkeep)	**6.75 %**
Out of reach: rents, energy, transport, schooling, catering	**45.53 %**

16.6 % to 54.5 %

depending on how food is counted. The three heaviest items are out of reach on their own: catering 13.27 %, rents 10.65 %, communication services 6.71 % — **30.6 %**.

Not a defect to fix: a marketplace sells goods, and close to half the basket is services and housing.

INS Côte d'Ivoire — IHPC publication, August 2026, table 6 · 50 groups, sum checked at 10,000

02

The concepts

Six words to have in mind before looking at any figure.



COICOP 2018 — the classification

The classification of individual consumption by purpose sorts every household outlay into **13 divisions**, then groups, classes and subclasses.

level	example	what it is for
division	04 — Housing, water, electricity, gas	the level of the published weights
group	04.1 — Actual rentals for housing	the level of this project's codes
class	04.1.1 — Rents paid by tenants	the level at which an index aggregates
subclass	04.1.1.0 — Main residence	the level of the basket's varieties
variety	04.1.1.0.10 — a room in a shack	what an enumerator actually records

TWO REVISIONS COEXIST

The WAEMU harmonised CPI — Côte d'Ivoire, Senegal — long followed **COICOP 1999 with 12 functions**; Côte d'Ivoire has moved to **2018 with 13 divisions**. The two do not overlap, and the project handles that explicitly.

United Nations, COICOP 2018 · classification supplied by INS Côte d'Ivoire (625 rows, 13 divisions)

Basket, variety, SKU — three identities

The basket

The list of what is recorded. For a statistical office: **971 varieties** in Côte d'Ivoire. For this device: **25 site categories** per country, chosen and written to a file.

The variety

An item defined precisely enough to be found again next month: « gross monthly rent of a 2-to-3-room house in a shared yard with toilets ».

The SKU

The seller's internal reference — « AR697FC05DC7BNAFAMZ ». Name, URL and price change: **the SKU is the only fixed point** between two collections.

6,762

matched SKUs

Why the SKU decides everything: an index compares **the same item** at two dates. In the Ivorian collection of 14 and 18 October, 6,762 SKUs appear at both dates — hence 6,762 comparable price pairs, of which **325 moved**.

Matched models and elementary formulas

An elementary index summarises the price movement of a group of items. Three formulas, three different answers.

formula	what it computes	behaviour
Jevons	**geometric mean of price ratios**	**the one retained: insensitive to outliers**
Dutot	ratio of mean prices	sensitive to a mix of expensive and cheap items
Carli	arithmetic mean of price ratios	known upward bias; computed in order to show it

THE MATCHED-MODEL RULE

An item enters the computation only if it is present at **both dates**. The project's safeguards: at least **3 matched items** per stratum, a gap of **2 periods** tolerated, and any price ratio outside **[1/3 ; 3]** discarded — a price that triples in a month is almost always a data-entry error, not inflation.

Weights — what turns an average into an index

A statistical office measures, through a household survey, how a budget is split. Out of 100 F spent in Côte d'Ivoire:

31.17 F on food products

13.27 F on catering

10.65 F on rent

country	food	housing	unit	sum check
Côte d'Ivoire	31.93 %	15.50 %	per ten thousand	10,000.00
Senegal	49.64 %	12.04 %	per ten thousand	10,000
Uganda	27.05 %	10.42 %	per thousand	999.9999

Without weights, an electricity tariff would count as much as a bag of rice. With them, we can say what the device covers — **the only figure that honestly says what it is worth**.

Every extraction is refused if the sum does not come out right: a wrong sum means the layout has changed and the figures read are not the right ones.

INS Côte d'Ivoire, IHPC August 2026 · ANSD Senegal, IHPC August 2024 · UBOS, CPI February 2026

robots.txt — the rule we do not work around

Every site publishes at its root a file saying what a program may read. The device respects it without exception.

situation	what the device does	why
a rule forbids the address	it does not go there	it is the publisher's decision
the file answers 404	it goes there	RFC 9309 makes that permission
the file is unreadable	it does not go there	RFC 9309 requires refusing everything
the site answers 403 to the robot	it does not insist	an anti-bot block is a decision, not an obstacle



THE DISTINCTION THAT COST A CAMPAIGN

A robots.txt that becomes **unreadable under load** is treated as a blanket refusal. Six countries out of eight were lost that way on one collection, and Nigeria a second time on rents — although its file allows the page. The log now names the nature of the refusal.

RFC 9309, Robots Exclusion Protocol · rfc-editor.org/rfc/rfc9309.html

03

The chain

From web sites to the index, in three notebooks and two families of sources.



The whole device in one diagram

1 · COLLECT

Notebook 1

8 Jumia sites
25 categories, 10 pages

→ **68,884 products**

2 · COMPLETE

Notebook 2

offices, regulators, WFP
+ 7 classifieds sites

→ **53,149 records**

3 · ANALYSE

Notebook 3

COICOP classification
indices and checks

→ **38 product rules**

WHAT TRAVELS BETWEEN THE NOTEBOOKS

`paniers/panier_XX.xlsx`	`outputs/<ISO3>/panel_XX_*.zip`	`outputs/<ISO3>/officiel_XX_*.zip`	`sorties/excel/*.xlsx`
the basket, reviewed and amendable	the collection archive, typed	records, indices, log	the same data, readable

Notebook 3 reads the **archives**, not the Excel workbooks: an archive's parquet is typed, whereas reading back from Excel turns division « 01 » into the integer 1.

Three kinds of output, kept apart on purpose

The records

prices

They aggregate together: a WFP market price and a Jumia price are both prices.

The indices

already built

Mixing them with prices would amount to computing an index of indices. They are joined at **division level**.

The log

what each source gave

Including when it gave nothing. **A silent source is information**, not silence.

MEASURED EXAMPLE

Senegal returns **9,624 records** and **340 index rows** — the latter come from ARTP, which publishes a mobile services index quarterly. Adding them up would make no sense.

04

The three notebooks

Purpose, sources and outputs of each — and classification in detail.



Collecting e-commerce

Recording, across eight marketplaces, the price of everything that enters a household's basket.

WHAT A ROW HOLDS

`sku`
the seller's reference, the matching key

`name`
the displayed label, basis of the classification

`price + currency`
the price and its currency, never assumed

`category_slug`
the category it came from

`collect_date`
the date of the record

what the collection does	how
discovers the site's category tree	menus and departments, two levels
picks 25 categories	one per COICOP division, in turn
reads 10 pages per category	2.5 s delay between requests
writes an archive and a workbook	nothing is ever overwritten

68,884

distinct products across 8 countries on 2026-10-28, from 29 categories in Egypt to 32 in Côte d'Ivoire.

jumia.ci · jumia.sn · jumia.com.ng · jumia.co.ke · jumia.com.gh · jumia.ug · jumia.com.eg · jumia.ma

Completing with official sources

E-commerce sells neither electricity, nor rent, nor fresh produce. Notebook 2 finds what is missing where it is published.

source	what it brings	division	country
WFP — HDX HAPI	market prices, field records	01	all 8
OCPV, Sikafinance	retail and wholesale prices	01	CI
ANARE-CI	regulated electricity tariffs	04	CI
CRSE	regulated tariffs	04	SN
ARTP — IPSM	mobile services index, quarterly	08	SN
7 classifieds sites	housing rents	04	all 8

Result: **53,149 records**, including 2,800 property listings (**2,295 recognised as rentals**) and 340 index rows.

A CONDITION OF THE SOURCE

HDX HAPI requires the caller to identify itself: the contact e-mail is not a project option but a condition of the source.

WFP via HDX HAPI · OCPV · Sikafinance · ANARE-CI · CRSE · ARTP (IPSM) · 7 property classifieds sites

Classify, compute, check

Notebook 3 collects nothing. It reads the archives, classifies every product, computes the indices and refuses to deliver a doubtful panel.

1 Load

the archives of notebooks 1 and 2, filtered to a single country: two currencies do not add up.

2 Classify

38 product rules and 7 category rules, multilingual embeddings and a cascade of language models. Every answer is kept.

3 Compute

elementary indices per stratum, then aggregation; Jevons, Dutot and Carli side by side.

4 Check

mixed currencies, strata too thin, ratios out of bounds: each one a reason to block.

A BLOCKING CHECK, NOT A WARNING

If the panel mixes two currencies, **utilisable** turns false and the index is not produced.

Classifying: rules, embeddings, models

Each method has a domain where it is best. The notebook measures them side by side; production chains them: **rules first, the model for what they cannot settle**.

Rules

38 patterns on the label, then on the department. Free, instant, reproducible — and **blind to what they did not foresee**.

Embeddings

A local multilingual model assigns the label to the nearest class. No quota, same result every time; **worth as much as its annotated references**.

Language model

Reads the whole label and returns class, quantity and brand as JSON. Costs money, has quotas, and **never answers twice the same way**.

Cascade

Groq, then Gemini, then Ollama, then the rules. A provider whose quota is reached is **set aside for the whole session**.

QUOTA REACHED (HTTP 429)

The client waits the delay the server asks for — at most 4 retries, never more than 60 s — then moves to the next provider. **A provider set aside does not come back**: before, every batch lost three calls asking it again.

The rules: keywords, read in order

1 Normalise
lower case, no accents:
« Réfrigérateur » =
« refrigerateur »

2 Search
in the label first, then label + Jumia department

3 First rule
that matches, from the most specific to the most general

4 Else the department
gives the division only: 05, 08, 13… no class

rule (COICOP class)	keywords (extract)	real label matched	word
01.1.1 Bread and cereals	riz · rice · pâtes · pasta · farine · flour	Riz Basmati 900G X20	`riz`
05.3 Household appliances	réfrigérateur · fridge · bouilloire · kettle	Bouilloire électrique 2 litres	`bouilloire`
08.1 Communication equipment	smartphone · laptop · tv · haut-parleur	Haut Parleur Bluetooth Karaoke 2 Micros	`haut parleur`
13.1 Personal care	shampo · savon · soap · dentifrice	Anti-Dandruff Shampoo (500ml) [NG]	`shampo`
03.2 Footwear	chaussures · shoes · sneakers · sandales	Sandales Nu-pieds pour Homme - Noir	`sandales`
06.1 Medicines and health	vitamine · vitamin · paracetamol	Vitamin D3 10,000 IU, 100 Softgels [NG]	`vitamin`
05.6 Household upkeep	lessive · detergent · javel · bleach	Détergent Nil Câlin Assouplissant	`detergent`

Every rule carries both vocabularies, French and English. Labels recorded on jumia.ci and jumia.com.ng, October 2026.

Where the rules go wrong, and what catches them

When a word is there for another reason: **the code given is wrong, and given with confidence**.

real label	word	code given by the rule	what it is
Machine à Pop Corn électrique (Grande)	`corn`	01.1.1 bread and cereals	household appliance, 05.3
NUBIANCE Paris Lait Corps Unifiant & Hydratant	`lait`	01.1.4 milk, cheese, eggs	body care, 13.1
Machine à Café Semi-Automatique Professionnelle	`cafe`	01.2.2 coffee	household appliance, 05.3
Durag En Satin De Soie, Rouge Vin	`vin`	02.1 alcoholic beverages	personal effect, 13.2
Rangement De 40 Capsules Nespresso	`capsules`	06.1 medicines	kitchen utensil, 05.4

NO KEYWORD: THE DEPARTMENT DECIDES

« Scie Circulaire 2200W » · DIY department → **05**, no class

« Hydraulic Trolley Jack » [EG] · Exterior Accessories → **07**

« African Print Playsuit » [NG] · Baby Girls Fashion → **13**, not 03

FULL CODE, RULES ONLY · 28 OCT 2026

92.4 %	89.9 %	82.7 %	75.8 %
CI	MA	EG	NG

the rest has only the department's division. English-language sites lose 10 to 15 points.

What catches them: the language model in mixed mode, and **the annotated sample per country — the only measure of the real error rate**.

The prompt sent to the model, as it leaves

EXCERPT, SYSTEM MESSAGE

```
You are a consumer price statistician. For each product sold online in Africa,
you return its COICOP 2018 class, its net content, the number of units in the…

Allowed codes (as fine as possible):
- 01.1.1: Cereals and cereal products
…
- "quantity" and "unit" describe the content of ONE unit (e.g. "Water 1.5L - P…
- Units: kg, g, l, ml, cl, or "unit" for counted objects (phone, nappy, sachet…).
- Dosages (1000mg), storage capacities (128GB), power ratings (350W) and scree…
…
Answer only in JSON: {"items": [{"id": "...", "coicop": "...", "quantity": 1.5,
"unit": "l", "pack_count": 6, "brand": "...", "confidence": 0.9}]} with one el…
```

WHAT IT CONTAINS

- a role: consumer price statistician
- the 35 allowed classes and 13 divisions, with their label
- quantity rules: « 1.5 L × 6 » is a content, « 128 GB » is not
- the answer format: JSON, one element per product, a confidence 0–1
- the same text for Groq, Gemini and Ollama, in the notebook's language

batches of 25 products · **≤ 600 per run** · **temperature 0** · **schema-validated answer**

2,788 characters in French, 2,624 in English

The providers, observed in the field

provider	what is needed	observed in the field, October 2026	what the notebook does
Groq	a key, free of charge	HTTP 429 from the third batch of 25 on gpt-oss-20b; the catalogue lost the llama-3.x on 26/10	waits the requested delay, then moves to Gemini
Gemini	an AI Studio key	2.5 models closed to new accounts (HTTP 404); the 3.5 answer; an occasional 503 « high demand »	the start-up trial adopts the replacement the API names
Ollama	a local server, or the Colab box	Colab: the installer required zstd; 5 to 10 min per session; qwen2.5:3b, first call 99 s on a T4	installs, starts, waits for the server, downloads the model, names the failing step
rules	**nothing**	**always there**	**last resort**



LISTED IS NOT ACCESSIBLE

The model menu is built at run time: the notebook reads each catalogue, sets aside what is not a dialogue model, **makes one trial call per candidate** and only offers those that answered. On 28/10: four models at Groq, two at Gemini, one at Ollama — and Gemini's 2.5 shown with the reason for their refusal.

Embeddings: weak alone, strong annotated

HOW THEY CLASSIFY

- 800 prototypes for 35 classes: the French label, the English one, and one per rule keyword
- each label joins the class whose prototype is nearest in cosine; the similarity is the confidence, the gap with the second class the margin
- annotated products can be added to the prototypes: a label close to an annotated example takes its code
- default model multilingual-e5-large, local, no quota; three others on the menu, down to the lightest for a CPU

TEST BENCH: 380 PRODUCTS, 29 CLASSES

method	class	division	products/s
rules	98.7 %	98.7 %	6,396
embeddings, zero-shot	54.2 %	72.9 %	32
embeddings + references	**94.2 %**	**97.4 %**	**34**

Zero-shot only recognises class labels. With annotated references, in cross-validation, the same method catches up with the rules: **it is worth as much as the annotated sample it is given**.

The rules score 99 % here because this set shares their vocabulary. Only an annotated sample per country measures a real catalogue.

Model measured: multilingual-e5-base · five-fold cross-validation

05

The calculations, by example

Each measure walked through on a real case, with the intent behind the formula.



Example 1 — an elementary Jevons index

Three Ivorian SKUs actually recorded at both dates, and the computation step by step.

SKU	14 October	18 October	ratio
AR697FC05DC7BNAFAMZ	5,850	3,180	0.5436
AP009EA0RQQ13NAFAMZ	109,000	87,200	0.8000
2N436CM03UMQGNAFAMZ	6,099	5,899	0.9672

THE COMPUTATION

```
Jevons = (0.5436 × 0.8000 × 0.9672)^(1/3)
       = 0.7452
```

−25.5 % over four days for this three-item stratum

THE INTENT

The geometric mean treats a halving and a doubling as symmetric. An arithmetic mean of the ratios — Carli — would give **0.7703**, an upward bias of **+2.5 points** on these three items alone.

jumia_CI_2026-10-18.xlsx, collections of 14 and 18 October 2026

Example 2 — from displayed to comparable price

Two packs of rice at the same price are not at the same price. Extracting the quantity makes the offers comparable.

label recorded	quantity read	unit	unit price
Mineral water 1.5 L — Pack of 6	9.0	litre	price ÷ 9
Noodles — Pack of 40 x 70 g	2.8	kilo	price ÷ 2.8
Broken rice — 25 kg bag	25.0	kilo	price ÷ 25
Vitamin C 1000 mg — 30 tablets	30.0	unit	price ÷ 30
Smartphone 6.6" 128 GB	1.0	unit	the price

THE INTENT, AND THE TRAP

A dosage of 1000 mg, a storage capacity of 128 GB, a power rating of 350 W or a screen diagonal are **not quantities**: reading them as such would give an absurd price per kilo. The unit price is therefore computed only where it means something — food, beverages, upkeep, fuels, personal care: **01, 02, 05.6, 07.2, 13.1**.

Example 3 — classifying, and the false friends

The classifier reads the label, then the category. The four cases below were genuinely misclassified before correction.

label recorded	wrongly read as	correct classification
Butterfly Baby Potty	butter — 01.1.5	childcare — 13.2
Car Cover, Fits To The Size Of The Car	tea — 01.2.3	car accessory — 07.2
Self-adhesive wallpaper, elegant	gloves and tights — 03.1	unclassified, for want of better
3pcs Milk Feeding Bottle Set	milk — 01.1.4	feeding bottles — 05.4 per the standard

THE LESSON

All these errors have the same shape: **a pattern with no word boundary**. « butter » caught « Butterfly », « gant » caught « élé-gant », « collant » caught « auto-collant », and the French « the » for tea caught the English article. Each case is now pinned by a named test.

Example 4 — matching a rent to its variety

Comparing a median of listings with a survey price makes no sense. Variety by variety, the comparison does.

	median of listings	INS survey price	gap
All listings together	850,000 FCFA	114,146 FCFA	× 7.45
Matched studio	**120,000 FCFA**	**80,639 FCFA**	**× 1.49**
Matched 2–3 room flat	**300,000 FCFA**	**114,146 FCFA**	**× 2.63**

×7.45 → ×1.5–2.6

gap before → after matching

97 %

1,105 of 1,145 listings matched

WHAT MATCHING SETTLES, AND WHAT IT LEAVES

The rest was not a measurement bias but a difference of housing stock — the sites advertise villas in Cocody, the official basket holds « a room in a shack ». What remains is the standing of the neighbourhood, which listings do not carry: **matching holds on two dimensions out of four**.

sorties/excel/appariement_loyers_CI.xlsx · INS Côte d'Ivoire, 20 varieties of subclass 04.1.1.0

06

Using and running it

What to install, what to type, and how to resume an interrupted collection.



The only real prerequisite

Python 3.10 or above

3.12.0 validated on a workstation. It is the only prerequisite: the dependencies install afterwards.

JupyterLab

To open the notebooks.

```
pip install jupyterlab
```

THEN, IN THE PROJECT FOLDER

`pip install -r requirements.txt`	the dependencies of the device
`jupyter lab`	opens the notebooks in the browser
`python -m pytest -q`	277 tests, no network

Launching a full campaign

Four commands, in this order. Each writes its results before moving to the next country.

1

Propose the baskets

writes paniers/panier_XX.xlsx — to be reviewed before going further

```
python scripts/exporter_paniers.py --contact you@institute.org
```

2

Collect e-commerce

several hours; a 240 s pause between two countries

```
python scripts/collecter_tous.py --contact … --max-pages 10
```

3

Collect the official sources

a few minutes per country; rents included

```
python scripts/collecter_officiel_tous.py --contact …
```

4

Analyse

the archives are found without typing anything

```
open notebooks/3_analysis_EN.ipynb, pick the country
```

Resuming, and never overwriting

situation	what happens
a collection is interrupted	the countries already done are written; the rest is relaunched
a country fails	its incident is logged, the others carry on
it is relaunched the same day	the file takes a _2, _3 suffix — nothing is replaced
a page has already been read	the cache returns it without a new request
a site refuses through robots.txt	the source is set aside and the reason written to the log



WHY NOTHING IS EVER OVERWRITTEN

A collection of eight thousand products was once lost on this project — not through a bug, but through a command that thought it was tidying up. Since then names carry the timestamp to the minute and writing stops if the file exists. The downside: **several passes coexist**, and only the last one is authoritative. python scripts/inventaire.py says which.

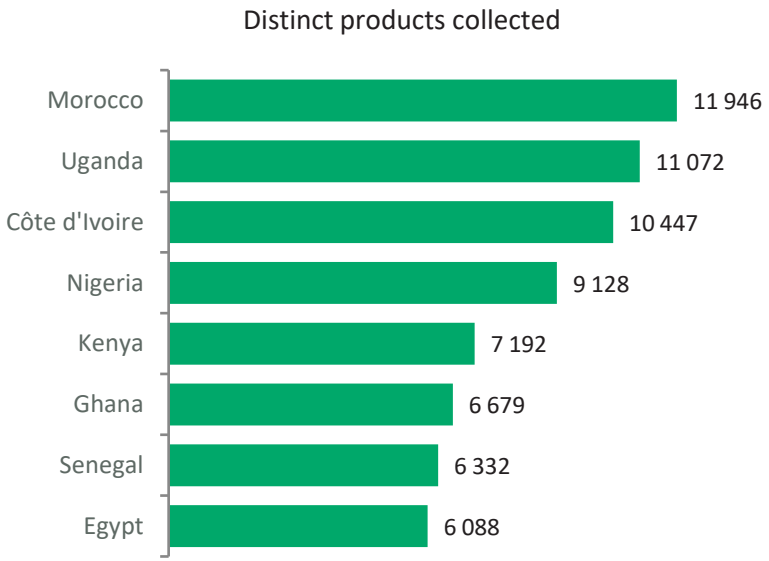
07

The data collected

What is on disk, country by country, at the minute this deck was produced.



E-commerce — notebook 1



country	site	categ.	currency	day
Morocco	jumia.ma	39	MAD	2026-10-20
Uganda	jumia.ug	42	UGX	2026-10-21
Côte d'Ivoire	jumia.ci	32	XOF	2026-10-21
Nigeria	jumia.com.ng	41	NGN	2026-10-20
Kenya	jumia.co.ke	30	KES	2026-10-21
Ghana	jumia.com.gh	37	GHS	2026-10-21
Senegal	jumia.sn	31	XOF	2026-10-21
Egypt	jumia.com.eg	29	EGP	2026-10-20

The number of categories is the figure to watch: it says how many departments the basket actually reached.

Total: **68,884** distinct products across 8 countries.

last collection per country · measured on 2026-10-28 12:10

Official sources and rents — notebook 2

country	records	of which rents	rentals	indices	sources
Nigeria	15,729	216	208	0	2
Senegal	9,624	326	284	340	4
Uganda	9,107	19	2	0	2
Ghana	6,916	166	166	0	2
Côte d'Ivoire	5,627	1,582	1,145	0	5
Kenya	5,417	209	209	0	2
Egypt	696	249	249	0	2
Morocco	33	33	32	0	1

53,149
records in total

2,295
rentals among 2,800 property listings

340
index rows, from Senegal's ARTP

The 340 index rows are published quarterly: **they do not add up with prices**. Only rentals enter the index: a sale price is an asset's price, not a consumption price.

last collection per country · measured on 2026-10-28 12:10

Baskets and official weights

country	catalogue	basket	divisions	weights	official
Morocco	1,329	25	8	0	—
Côte d'Ivoire	**1,098**	**25**	**8**	**25**	**yes**
Kenya	958	25	9	3	—
Ghana	947	25	8	4	—
Nigeria	840	25	9	3	—
Senegal	**666**	**25**	**8**	**25**	**yes**
Uganda	**467**	**25**	**8**	**25**	**yes**
Egypt	417	25	8	0	—

1,329 → 25

categories offered in Morocco → categories recorded. The catalogue is everything the site offers; the basket is what we record.

OFFICIAL WEIGHTS, CHECKED

- ✓ **Côte d'Ivoire**
- ✓ **Senegal**
- ✓ **Uganda**

paniers/panier_XX.xlsx · paniers/ponderations_*.csv

08

Scope and limits

What the device brings, and the five places where it stops.



What the device brings

Cost

A campaign across eight countries fits in a few machine-hours. No enumerator, no travel.

Frequency

Nothing prevents a weekly collection, where a field survey is monthly.

Traceability

Every price carries its source, its address and its date; every classification carries the word that triggered it.

Reproducibility

An archive, a log, 277 tests. A third party can redo the computation.

Complementarity

Official sources cover what the web does not sell: regulated tariffs, market prices, rents.

Honesty about the figure

The device can say what share of the official basket it observes — and that is rare.

The five limits, quantified

limit	measured extent	consequence
The basket does not cover everything	16.6 % to 54.5 % of the Ivorian weights	this is not a CPI
Listings are asking prices	× 1.49 to × 2.63 after matching	a complement, not a substitute
The advertised stock is not the inhabited stock	× 7.45 before matching	matching is mandatory
Categories are chosen, not weighted	25 out of 417 to 1,329 depending on the country	human review is required
Rule-based classification gets things wrong	38 rules, 4 false friends fixed	the official exclusions table would do better



WHAT THESE LIMITS ARE NOT

They are not implementation defects that more effort would lift. A marketplace **sells goods**: rents, energy, transport, schooling and catering — close to half the basket — escape it by nature. The device is useful because it measures that share instead of passing over it.

The data sources, with their address

source	what it gives	address
Jumia, 8 countries	e-commerce prices	jumia.ci · .sn · .com.ng · .co.ke · .com.gh · .ug · .com.eg · .ma
WFP via HDX HAPI	market prices, 8 countries	hapi.humdata.org/api/v2
OCPV	retail prices, Côte d'Ivoire	infoprixocpv.com
ANARE-CI	electricity tariffs	anare.ci
CRSE	regulated tariffs, Senegal	crse.sn
ARTP	mobile services index	artp.sn/indice-des-prix-des-services-mobiles-ispm
Property classifieds	rents, 8 countries	coinafrique.com · propertypro.ng · property24.co.ke · meqasa.com · realestatedatabase.net · propertyfinder.eg · mubawab.ma

The method and classification sources

document	what it provides	address or origin
COICOP 2018, United Nations	the classification, 13 divisions	workbook supplied by INS Côte d'Ivoire, 625 rows
Panier CI, INS Côte d'Ivoire	971 official varieties, 20 of them rents	« COICOP Complète.xlsx », sheet « Panier CI »
Products and exclusions table	1,196 subclasses with their exclusions	same workbook, sheet « COICOP Prod&Exclusion »
INS Côte d'Ivoire, IHPC August 2026	weights, 50 groups, sum 10,000	« 8-PUBLICATION IHPC AOUT 2026.xlsx », sheet Page 5
ANSD Senegal, IHPC August 2024	12 functions and 48 groups, sum 10,000	ansd.sn/sites/default/files/2024-09/IHPC_08_2024.pdf
UBOS, CPI February 2026	13 COICOP divisions, sum 1,000	ubos.org/wp-content/uploads/publications/
RFC 9309	the robots.txt rule	rfc-editor.org/rfc/rfc9309.html

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What remains to be done

01	the official weights of the five missing countries — the method is proven on three
02	the neighbourhood → standing table, to complete the rent matching
03	an annotated reference set with ANSTAT: the first African COICOP benchmark, and the arbiter of the three methods
04	a second collection wave: without two dates, no index

